EXPERIMENT 6

Write the Python program for the Vacuum Cleaner problem

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Aim

To write a Python program that simulates a vacuum cleaner agent in a two-room world (Room A and Room B), which senses its location and the status of the room (clean or dirty) and chooses actions so that both rooms become clean, using a simple reflex agent.

Algorithm

Step 1: Define the environment with two rooms, A and B. Each room is either Clean or Dirty. The agent starts in one of the rooms.

Step 2: The percept of the agent is a pair (location, status), i.e. where it is and whether that room is Dirty or Clean.

Step 3: Define the agent function (simple reflex rules): if the status is Dirty, the action is Suck; else if the location is A, the action is Right; else the action is Left.

Step 4: Read the percept for the current location and obtain the action from the agent function.

Step 5: Perform the action: Suck makes the current room Clean; Right moves the agent to room B; Left moves the agent to room A.

Step 6: Display the step number, location, percept, action and the status of both rooms.

Step 7: Repeat from step 4 until both rooms are Clean (or the step limit is reached). Print the number of rooms cleaned and the number of moves made.

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Program

```
# Vacuum Cleaner problem: two rooms (A and B), simple reflex agent

def reflex_vacuum_agent(location, status):
    """Percept = (location, status of that room). Returns an action."""
    if status == "Dirty":
        return "Suck"
    elif location == "A":
        return "Right"
    else:
        return "Left"

def run(location, world, max_steps=10):
    print(f"Initial state: Agent in {location}, Room A = {world['A']}, Room B = {world['B']}\n")
    print(f"{'Step':<6}{'Location':<10}{'Percept':<18}{'Action':<8}{'Room A':<8}{'Room B':<8}")
    print("-" * 58)
    moves = cleaned = 0
    for step in range(1, max_steps + 1):
        status = world[location]
        action = reflex_vacuum_agent(location, status)
        percept = f"({location}, {status})"
        before = location
        if action == "Suck":
            world[location] = "Clean"
            cleaned += 1
        elif action == "Right":
            location = "B"
            moves += 1
        else:
            location = "A"
            moves += 1
        print(f"{step:<6}{before:<10}{percept:<18}{action:<8}{world['A']:<8}{world['B']:<8}")
        if world["A"] == "Clean" and world["B"] == "Clean":
            print("\nBoth rooms are clean. Goal reached.")
            break
    print(f"Rooms cleaned: {cleaned}, Moves made: {moves}, Total actions: {cleaned + moves}")

if __name__ == "__main__":
    run("A", {"A": "Dirty", "B": "Dirty"})
    print("\n" + "=" * 58 + "\n")
    run("B", {"A": "Dirty", "B": "Clean"})
```

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Output

```
Initial state: Agent in A, Room A = Dirty, Room B = Dirty

Step  Location  Percept           Action  Room A  Room B
------------------------------------------------------------
1     A         (A, Dirty)        Suck    Clean   Dirty
2     A         (A, Clean)        Right   Clean   Dirty
3     B         (B, Dirty)        Suck    Clean   Clean

Both rooms are clean. Goal reached.
Rooms cleaned: 2, Moves made: 1, Total actions: 3

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Initial state: Agent in B, Room A = Dirty, Room B = Clean

Step  Location  Percept           Action  Room A  Room B
------------------------------------------------------------
1     B         (B, Clean)        Left    Dirty   Clean
2     A         (A, Dirty)        Suck    Clean   Clean

Both rooms are clean. Goal reached.
Rooms cleaned: 1, Moves made: 1, Total actions: 2
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